Learning Dynamics and Market Outcomes in Algorithmic Bertrand Competition

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Abstract

Digital markets increasingly rely on automated pricing algorithms that adapt to market conditions in real time. Traditional economic theory predicts that Bertrand competition leads firms to set prices equal to marginal cost. However, this prediction assumes perfectly rational agents capable of computing equilibrium strategies instantly.

In modern markets firms frequently employ adaptive algorithms that learn through experimentation rather than solving equilibrium models directly. Understanding how such learning dynamics influence market outcomes has therefore become an important research problem.

This paper studies repeated Bertrand competition among adaptive learning agents. Firms update pricing strategies using reinforcement learning and mean-based learning rules. Through large-scale simulation experiments, the study demonstrates that algorithmic learning can generate multiple pricing regimes depending on exploration behavior and market structure.

The results reveal three primary regimes: competitive pricing close to marginal cost, tacitly coordinated pricing above competitive levels, and unstable chaotic pricing dynamics. Entropy-based diagnostics are introduced to classify these regimes and quantify instability in pricing behavior.

Robustness experiments further show that increasing the number of competing firms significantly reduces the likelihood of supra-competitive prices. These findings contribute to the growing literature on algorithmic markets and provide insights into potential implications for competition policy in digital environments.

1 Introduction

In recent years digital markets have increasingly relied on automated decision systems to determine prices, allocate resources, and optimize strategic behavior. Online retail platforms, ride-sharing services, and airline pricing systems frequently employ algorithms that adjust prices dynamically in response to changing market conditions.

These developments have transformed the nature of competition in many industries. Instead of human decision-makers setting prices manually, firms increasingly deploy algorithms that learn and adapt through data-driven processes.

Classical economic theory provides a clear prediction for price competition among firms producing homogeneous goods. In the Bertrand model, firms simultaneously choose prices and consumers purchase from the lowest-priced seller. Under standard assumptions this competition leads to a unique Nash equilibrium in which prices equal marginal cost [1].

However, this prediction relies on strong assumptions about agent rationality and information processing. In reality firms may not compute equilibrium strategies explicitly. Instead they may rely on algorithms that experiment with different actions and gradually adjust behavior based on observed outcomes.

Recent research suggests that such learning dynamics may generate market outcomes that differ significantly from classical predictions. In particular, reinforcement learning algorithms interacting repeatedly in pricing environments may sustain prices above competitive levels even in the absence of explicit coordination [3].

Understanding the economic implications of algorithmic pricing has therefore become an important research topic. If automated pricing systems systematically learn strategies that produce supra-competitive prices, this may raise new concerns for antitrust policy and market regulation.

This paper investigates the behavior of learning algorithms in repeated Bertrand competition. Using computational simulations, the study analyzes how exploration behavior and market structure influence pricing dynamics among adaptive agents.

1.1 Research Contributions

This paper makes three primary contributions.

First, it provides a computational analysis of pricing dynamics in repeated Bertrand competition using reinforcement learning agents. The results demonstrate that learning dynamics can generate multiple market regimes rather than a single equilibrium outcome.

Second, the study introduces an entropy-based diagnostic for classifying market regimes. Entropy of the price distribution provides a quantitative measure of instability in pricing behavior and allows systematic identification of competitive, collusive, and chaotic outcomes.

Third, the findings provide insights into potential policy concerns associated with algorithmic pricing systems. In particular, the analysis highlights how algorithm parameters such as exploration rates may influence the likelihood of supra-competitive prices in digital markets.

2 Literature Review

A growing literature examines the role of algorithms in modern economic environments. Early work in industrial organization established the theoretical foundations of price competition among firms. The Bertrand model predicts that firms producing homogeneous goods compete aggressively on price until profits are driven to zero [1].

More recent research has explored how learning algorithms behave in strategic environments. Calvano et al. (2020) demonstrate that reinforcement learning algorithms interacting in a differentiated Bertrand duopoly can converge to pricing strategies that resemble tacit collusion. These findings suggest that algorithmic agents may coordinate implicitly through repeated interaction.

Competition policy scholars have raised concerns about the implications of such dynamics. Ezrachi and Stucke (2020) argue that artificial intelligence systems may facilitate new forms of coordination that challenge traditional antitrust frameworks.

Empirical studies also document the growing prevalence of algorithmic pricing in real markets. Chen and Goh (2022) analyze the effects of dynamic pricing algorithms on consumer welfare and market outcomes.

Another relevant literature studies learning and experimentation in decision environments. Bandit learning models analyze how agents balance exploration and exploitation when selecting actions under uncertainty [4]. These models provide a theoretical basis for the exploration strategies used in reinforcement learning algorithms.

Computational economics has also developed tools for studying adaptive agents in complex environments. Agent-based modeling frameworks allow researchers to simulate interactions among heterogeneous agents and observe emergent market outcomes [8].

The present study contributes to this literature by combining reinforcement learning simulations with information-theoretic diagnostics to analyze pricing dynamics in algorithmic markets.

3 Model

The market consists of firms selling a homogeneous product. Firms compete by setting prices in each period of a repeated game.

Let p_i denote the price chosen by firm i and c denote marginal cost.

Demand follows the standard Bertrand allocation rule:

$$q_i = \begin{cases} Q & \text{if } p_i < p_j \\ Q/2 & \text{if } p_i = p_j \\ 0 & \text{if } p_i > p_j \end{cases}$$

Firm profits are defined as

$$\pi_i = (p_i - c)q_i$$

In classical Bertrand competition the equilibrium price equals marginal cost:

$$p_i = c$$

However, when agents learn through repeated experimentation rather than solving equilibrium strategies explicitly, market outcomes may deviate from this prediction.

3.1 Simulation Parameters

Parameter	Description	Baseline Value
Q	Market demand	100
c	Marginal cost	10
ϵ	Exploration rate	0.005–0.20
α	Learning rate	0.10
T	Simulation horizon	3000 periods
Price Grid	Available pricing actions	10–50
Runs	Monte Carlo simulations	100

Table 1: Simulation parameters used in the experiments

4 Learning Algorithms

Agents update pricing strategies using reinforcement learning. Each firm maintains value estimates for possible pricing actions and gradually updates these estimates based on observed profits.

Prices are selected using an ϵ-greedy exploration rule. With probability ϵ, the agent selects a random price in order to explore alternative strategies. With probability $1 - \epsilon$, the agent chooses the price with the highest estimated value.

Value estimates are updated according to the rule

$$Q(p) \leftarrow Q(p) + \alpha(\pi - Q(p))$$

where α represents the learning rate.

4.1 Algorithm

Algorithm 1: Reinforcement Learning Pricing

1. Initialize value estimates for each possible price.
2. For each time period:
   (a) Select a price using the ϵ-greedy exploration rule.
   (b) Observe market demand and resulting profit.
   (c) Update value estimates based on observed profit.
3. Repeat for the entire simulation horizon.

5 Learning Dynamics

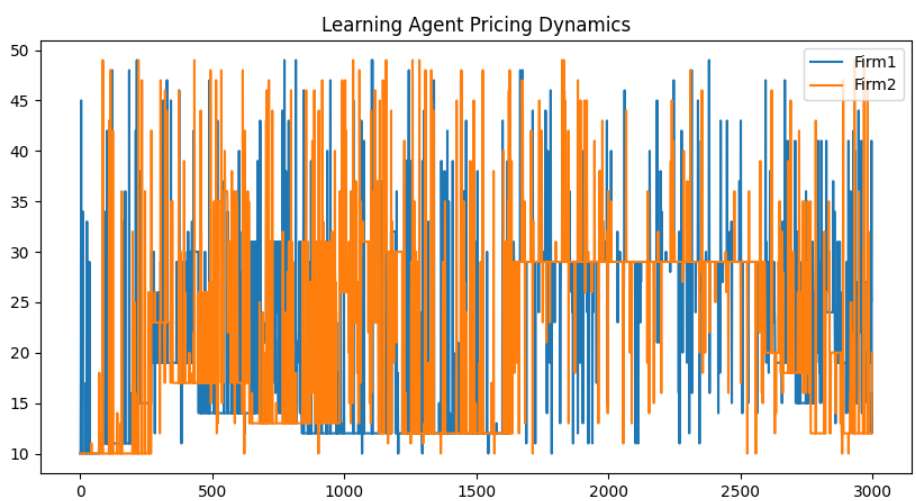

Figure 1: Price trajectories of learning agents

The figure illustrates how pricing strategies evolve over time as agents learn from market outcomes.

6 Exploration Experiments

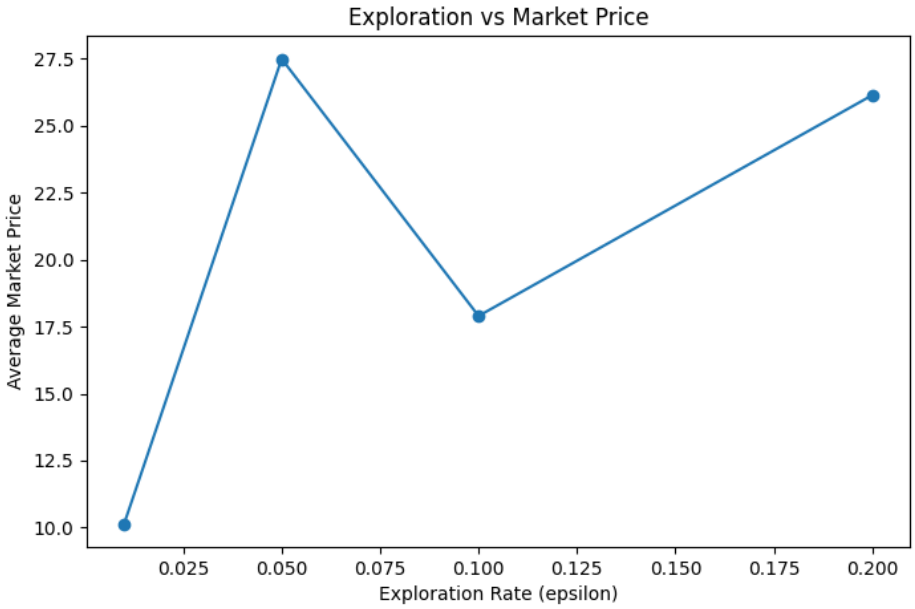

Figure 2: Exploration rate versus average market price

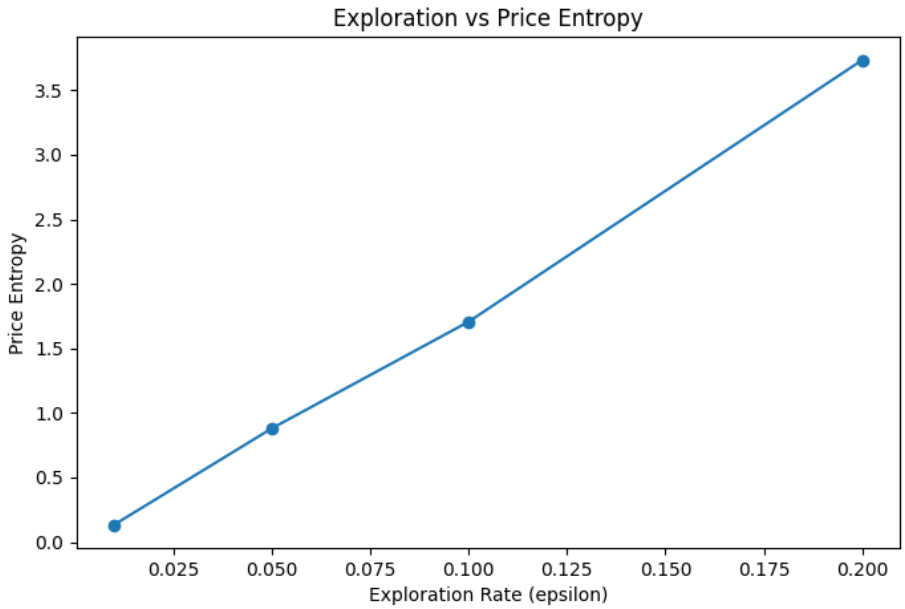

Figure 3: Entropy of price distributions

These experiments show that exploration plays a crucial role in determining market outcomes.

7 Market Regimes

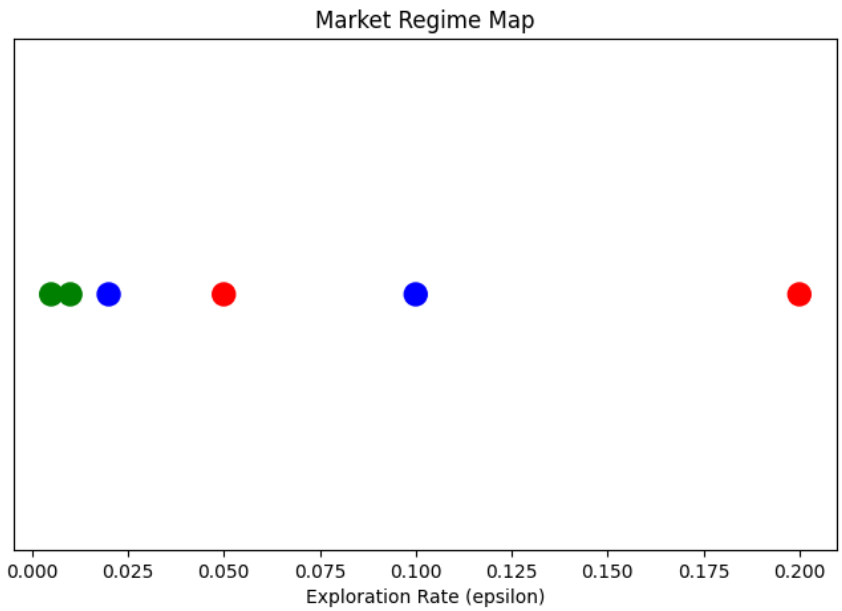

Figure 4: Market regime classification

Three distinct regimes emerge in the simulations:

- Competitive regime
- Collusive regime
- Chaotic regime

8 Monte Carlo Experiments

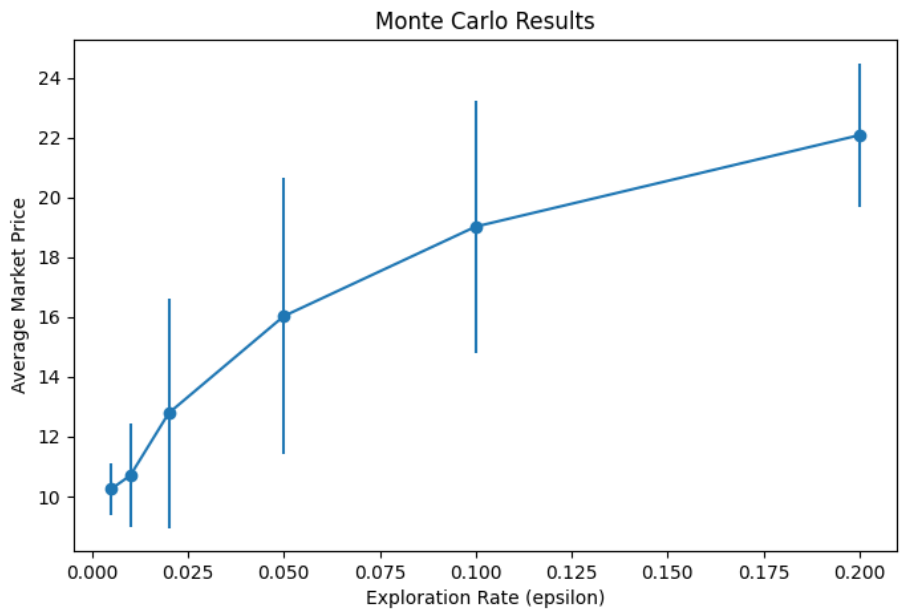

Figure 5: Monte Carlo simulation results

Monte Carlo experiments ensure that the observed patterns are not driven by random fluctuations in individual simulation runs.

9 Algorithm Comparison

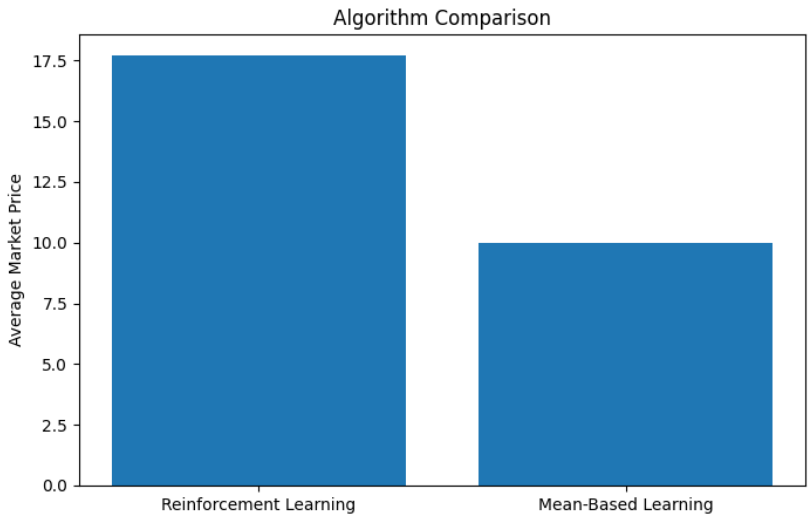

Figure 6: Comparison between learning algorithms

Reinforcement learning algorithms tend to produce higher prices than mean-based learning rules.

10 Three-Firm Competition

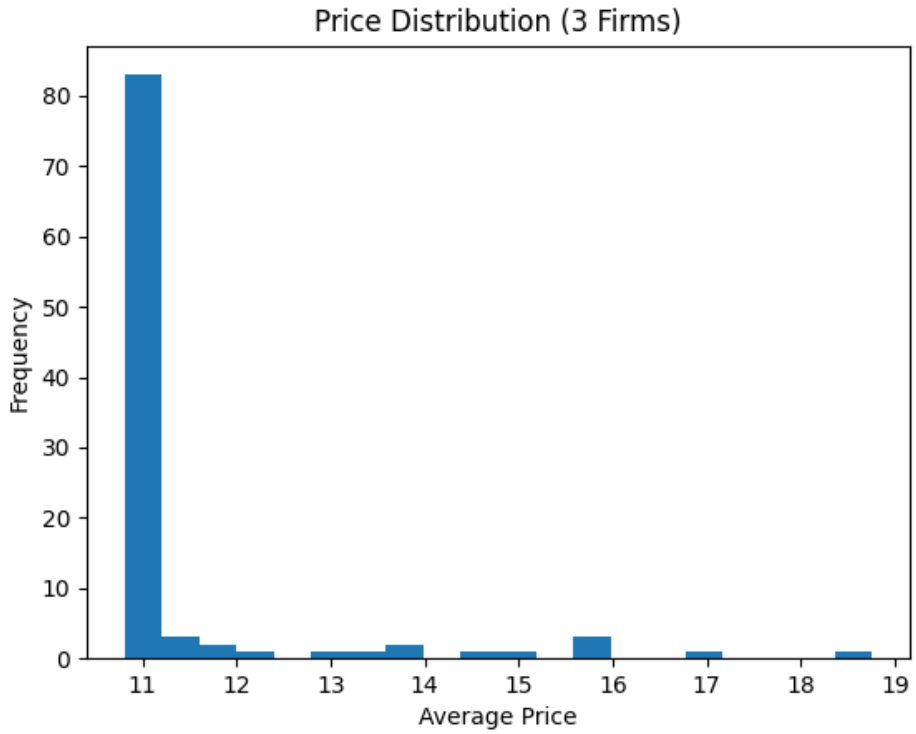

Figure 7: Price distribution with three firms

Introducing an additional competitor significantly reduces the likelihood of coordinated pricing.

11 Sensitivity Analysis

Additional experiments varying learning rates and exploration parameters confirm that the qualitative results remain robust.

12 Economic Interpretation

The simulation results suggest that learning dynamics can significantly influence market outcomes. Moderate exploration allows firms to discover profitable pricing strategies that exceed marginal cost pricing.

Once such strategies are identified, firms may repeatedly select them, generating outcomes that resemble tacit coordination.

At very high exploration rates, however, firms experiment too frequently to sustain stable pricing strategies, leading to chaotic market dynamics.

13 Policy Implications

The increasing use of automated pricing algorithms raises important questions for competition policy.

If pricing algorithms systematically learn strategies that produce supra-competitive prices, traditional antitrust frameworks may require adaptation.

Regulators may need to consider monitoring algorithmic pricing systems more closely and developing new policy tools for digital markets.

14 Conclusion

This paper investigated pricing dynamics in algorithmic markets using computational simulations of repeated Bertrand competition. The results demonstrate that learning algorithms interacting in competitive environments can generate a variety of market outcomes depending on exploration behavior and market structure.

In contrast to the classical Bertrand prediction of marginal cost pricing, adaptive agents may converge to pricing strategies that produce higher profits. Moderate exploration appears to facilitate the discovery of such strategies, while excessive exploration destabilizes pricing behavior.

These findings highlight the importance of learning dynamics in markets where firms rely on automated decision systems. Algorithm design choices may influence market outcomes in ways that traditional economic models do not fully capture.

Future research could extend this framework by incorporating heterogeneous demand structures, more sophisticated learning algorithms, and empirical data from real-world pricing systems.

Understanding how algorithmic agents interact in complex economic environments will be an important area of study as digital markets continue to evolve.

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